Sudeera Dilshan

AI/ML Engineer | AI Graduate | University of Moratuwa

sudeera-dilshan | SudeeraDilshan | @dilshanrgs31 | dilshanrgs31@gmail.com | +94 77 943 5845

About Me

B.Sc. (Hons) Artificial Intelligence graduate from the University of Moratuwa with hands-on industry experience in LLM/VLM integrations, agentic AI workflows, and MCP server development. Passionate about Data Science, machine learning, computer vision, and building production-ready AI systems. Skilled in full-stack and mobile app development, with a proven track record of delivering impactful AI solutions. A published researcher, technical content creator on Medium, and competitive programmer committed to continuous learning and responsible AI innovation.

Work Experience

Associate AI Engineer | Ariso Technologies (Pvt) Ltd **Aug 2025 – Oct 2025**

- Developed and integrated production-level LLM pipelines and MCP server components to power intelligent, context-aware AI services.
- Contributed to time-series forecasting model development and optimised deployed AI systems for scalability and reliability in production environments.
- Technologies: Python, LLMs, LangGraph, MCP, FastAPI, Docker, Prophet, statsmodels.

AI Engineer Intern | Ariso Technologies (Pvt) Ltd **Feb 2025 – Jul 2025**

- Built and deployed agentic AI workflows with multi-step reasoning and tool-use capabilities, integrating LLMs into real-world business automation pipelines.
- Engineered MCP server modules and AI-powered microservices, enabling seamless orchestration of external data sources and APIs.
- Designed and implemented an agent orchestration framework using Mosquitto message broker for sending commands to agents and Redis for distributed cache memory management.
- Built knowledge base services for LangChain and LangGraph agents, enabling dynamic assignment of relevant knowledge bases to specialised agents for context-aware reasoning.
- Created and managed agent workflows using Langflow, streamlining the visual design and orchestration of multi-agent AI pipelines.
- Technologies: Python, LangChain, LangGraph, LLMs, MCP, Langflow, Mosquitto, Redis, FastAPI, Docker.

Education

2022 – 2026 University of Moratuwa Colombo, Sri Lanka
B.Sc. (Hons) in Artificial Intelligence **GPA: 3.33**

2017 – 2019 Richmond College Galle, Sri Lanka
G.C.E. Advanced Level: Combined Mathematics – A | Physics – A | ICT – B **Z-Score: 2.2305**

Projects

Soft Prompting-Based SLM Generalization – Meta-Learning & Fuzzy Reasoning GitHub | *Research*
Research to improve cross-task generalisation in Small Language Models through chain-of-thought knowledge distillation and meta-learning. Designed a custom architecture with a lightweight MLP and a top fuzzy layer for adaptive soft prompt optimisation, and implemented a custom transformer, MLP, and ANFIS module with soft prompts trained using First-Order MAML (FOMAML) on Qwen2.5-0.5B.

Voicense – AI-Powered Lecture Note Generation Platform GitHub
Developed an AI-powered mobile application for automated lecture recording, transcription, and intelligent note generation, serving on the core backend and AI pipeline team. Delivered real-time transcription, structured notes, topic-wise summaries, and an AI chat assistant, with an asynchronous backend integrating Whisper AI and LangChain-based retrieval to ground responses in lecture content (Flutter, FastAPI, MongoDB, Gemini API, Docker).

Time Series Transit Forecast: Visualized Analysis GitHub
Automated a Socrata API data pipeline to extract and preprocess 20+ years (9,000+ records) of daily public transit data. Trained an XGBoost model on engineered temporal lag features that outperformed a Facebook Prophet baseline, and built an interactive Streamlit and Plotly app to explore trends, seasonal anomalies, and real-time forecasts.

Real-time Vehicle Speed Detection System GitHub
Built a real-time computer vision pipeline using YOLOv8 and OpenCV to detect, track, and estimate vehicle speeds on highways. Engineered a custom centroid tracking algorithm and spatial mapping with virtual reference lines, delivering directional vehicle counting and processed video output with bounding boxes and speed labels.

Bias-Free Loan Eligibility & Risk Explainability System GitHub
Built an end-to-end ML system for loan eligibility prediction with SHAP-based explainability and fairness auditing for transparent, ethical financial decisions. Integrated a small language model (Qwen2.5-0.5B) to turn SHAP explanations into human-readable narratives (XGBoost, FairLearn, FastAPI, ReactJS, Docker).

StreamAI – Hybrid Machine Learning Recommender System GitHub
Developed a hybrid recommendation engine combining K-Means for user segmentation, SVD (matrix factorization) for collaborative filtering, and TF-IDF for content-based filtering. Designed a dynamic routing algorithm that moves users from popularity-based to collaborative recommendations as their data profile grows, solving the cold-start problem, and packaged the pipeline into an interactive Streamlit dashboard (scikit-learn, scikit-surprise, Pandas).

Customer Support Bot on Fine-tuned SLM GitHub
Built an end-to-end pipeline to fine-tune Qwen2.5-0.5B-Instruct for customer support on a 4 GB VRAM GPU using QLoRA (4-bit quantization), LoRA adapters, micro-batching, and gradient accumulation. Converted 5,000 rows of the Bitext customer support dataset into ChatML format, trained with TRL's SFTTrainer, and delivered an interactive chat interface for the adapters (PyTorch, Transformers, PEFT, bitsandbytes).

Virtual Mouse Using Hand Gestures GitHub
Developed a hands-free virtual mouse using real-time webcam hand gesture recognition. Implemented hand landmark detection and tracking to map finger positions and gestures to cursor movement, clicks, scrolling, and drag actions (Python, OpenCV, MediaPipe, PyAutoGUI).

Publications

Voicense: AI-Powered Lecture Note Generation Tool Dec 2024
9th International Conference on Information Technology Research (ICITR) 2024, pp. 1–6, IEEE, University of Moratuwa, Sri Lanka. DOI: 10.1109/ICITR64794.2024.10857774

Technical Skills

Programming Languages	Python, Java, C#, R, C, Dart, JavaScript, SQL
GenAI & Agentic Frameworks	LangChain, LangGraph, Langflow, LangSmith, MCP
AI/ML Libraries	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, SHAP, HuggingFace, OpenCV, YOLOv8
Data Analysis	pandas, numpy, matplotlib, seaborn, Tableau
Big Data & Analytics	Apache Spark, Databricks
Data & Vector Stores	MongoDB, MySQL, PostgreSQL, Neo4j, FAISS, ChromaDB, Redis
MLOps, Cloud & Messaging	MLflow, Docker, AWS, Jenkins, GitHub Actions, Firebase, Mosquitto
Backend & Full-Stack	FastAPI, Django, Flask, NodeJS, ReactJS, Spring Boot, Flutter, Microservices
Version Control	Git

Certifications

Advanced Python Programming – Source Code
MCP: Build Rich-Context AI Apps with Anthropic – DeepLearning.AI
LangChain for LLM Application Development – DeepLearning.AI
Neo4j Certified Professional – Neo4j GraphAcademy
Kubernetes From 10000 Feet – CoDeKu DevOps Academy

Achievements & Extracurricular Activities

- **6th Place**, CodeRush 2023 (Organised by INTECS)
- **Participant**, ACES Coders V10.0 2023 – University of Peradeniya
- University of Moratuwa Karate Team Representative (2022, 2023)
- Technical Content Creator on Medium – @dilshanrgs31

References

Dr. (Ms.) I. Thilini S. Piyatilake
Head of Department
Department of Computational Mathematics
University of Moratuwa
thilinisp@uom.lk

Mr. Charith Bimsara
Associate Technical Lead – AI
Rootcode Labs
charith.bimsara@rootcodelabs.com